STA2002 HW3

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1 Q2.c

An estimator is consistent if it converges in probability to the true parameter value as $n \to \infty$. For $\hat{\lambda} = \frac{1}{\bar{X}}$ to be consistent, $\bar{X}$ must converge in probability to $E(X)$.

Since $X_i \sim Ga\,(a_0, b_0)$, by the law of large numbers, $\bar{X} \xrightarrow{P} E(X) = \frac{a_0}{b_0}$ as $n \to \infty$. Thus, we have:

$$\hat{\lambda} = \frac{1}{\bar{X}} \xrightarrow{P} \frac{b_0}{a_0}$$

This shows that $\hat{\lambda}$ is a consistent estimator for $\lambda_0 = \frac{b_0}{a_0}$ when the model is mis-specified(i.e. using the exponential model). Mis-specified means that we assume $X_i \sim \text{Exp}(\lambda)$, so we are essentially modeling the data as a Gamma distribution with shape parameter $a_0 = 1$ and rate parameter $b_0 = \lambda$. That is to say, $X_i \sim \text{Exp}\,(b_0)$, and $\hat{\lambda} \xrightarrow{P} b_0$, which is the correct rate parameter.

However, if $a_0 \neq 1$, then the underlying distribution is not exponential. $\hat{\lambda}$ is estimating a parameter $\left(\frac{b_0}{a_0}\right)$ that is different from the true rate parameter b_0.

- If $a_0 > 1$: $\hat{\lambda} \xrightarrow{P} b_0/a_0 < b_0$. The estimator underestimates the rate parameter.
- If $a_0 < 1$: $\hat{\lambda} \xrightarrow{P} b_0/a_0 > b_0$. The estimator overestimates the rate parameter

The magnitude of error is $|b_0/a_0 - b_0| = b_0\,|1/a_0 - 1|$, so the further a_0 is from 1 , the larger the misspecification error. The error scales linearly with b_0.

In summary, $\hat{\lambda}$ is consistent for b_0/a_0. However, this is only the "true" parameter when $a_0 = 1$, which is when Gamma reduces to Exponential. When the shape parameter a_0 is misspecified (i.e., $a_0 \neq 1$), the estimator $\hat{\lambda}$ converges in probability to $\frac{b_0}{a_0}$ rather than the true parameter b_0.